

DEMAND PREDICTION OF AGRICULTURAL CROPS USING ARTIFICIAL INTELLIGENCE

A PROJECT REPORT

Submitted in partial fulfillment of the requirements for the award of degree of

BACHELOR OF ENGINEERING / MASTER OF SCIENCE

IN

COMPUTER SCIENCE & ENGINEERING

Submitted By:

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Dept. of Computer Science & Engineering

Academic Year: 2025 – 2026

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CANDIDATE'S CERTIFICATE

This is to certify that the project work entitled '**Demand Prediction of Agricultural Crops Using Artificial Intelligence**' is a bona fide record of independent research and development work carried out by the candidate in partial fulfillment of the requirements for the award of the degree in Computer Science and Engineering during the academic year 2025–2026.

The project embodies the results of original investigations conducted by the student under institutional supervision. The algorithms, data structures, machine learning pipelines, and user interfaces described in this report have been tested, verified, and found to meet the academic standards and technical specifications required by the department.

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ACKNOWLEDGMENT

The completion of this comprehensive project report on agricultural demand prediction and decision support systems has been an enriching intellectual journey. We take this opportunity to express our profound sense of gratitude to all those who provided guidance, inspiration, and technical assistance throughout this work.

First and foremost, we express our heartfelt gratitude to our respected **Project Guide** for his invaluable guidance, constant encouragement, and insightful suggestions in shaping the machine learning architecture and mathematical formulations of this platform.

We are equally grateful to the **Head of Department**, Computer Science & Engineering, for providing the necessary computational laboratory facilities and creating a supportive research environment.

We extend our sincere thanks to the teaching and technical staff of the department for their continuous cooperation. We also acknowledge the open-source scientific Python community and the developers of Scikit-Learn, Flask, Chart.js, and Leaflet.js for making high-performance software libraries accessible.

Lastly, we thank our parents and fellow students whose moral support and constructive feedback served as a constant source of motivation.

Project Author & Team

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ABSTRACT

Agriculture forms the primary economic lifeline for over 54% of India's population. However, smallholder farmers face severe structural vulnerabilities, including uncoordinated planting that triggers disastrous market gluts, unscientific fertilizer application leading to soil degradation, volatile commodity price fluctuations at regional mandis, and crop damage from unmonitored leaf pathologies.

This project presents **Hanu Agri**, an end-to-end intelligent agricultural decision support and demand prediction system. The platform unifies eight supervised machine learning models, statistical econometrics, and agro-climatic algorithms into a responsive web application: (1) **Random Forest Crop Recommendation** achieving 99.4% accuracy based on soil N-P-K, pH, temperature, humidity, and rainfall; (2) **Gradient Boosting Regressor Mandi Price Forecasting** with harmonic cyclical sinusoidal date embeddings; (3) **Decision Tree Fertilizer Advisory** providing balanced chemical and organic dosages; (4) **Demand-Supply Glut Risk Analyzer** calculating overproduction risk ratios; (5) **Haversine Geodesic Distance Engine** linking farmers to nearby APMC wholesale markets; (6) **Hargreaves-Samani Evapotranspiration Model** calculating daily crop water needs in Liters/Acre; (7) **Farm Budgeting & ROI Valuation Calculator** estimating net profit margins; and (8) **Multilingual Voice Conversational Bot** operating in English, Kannada, and Hindi.

Extensive empirical testing confirms sub-15 millisecond API inference latency and high predictive accuracy ($R^2 = 0.9412$ for price forecasting), establishing Hanu Agri as a robust, production-ready solution for modern precision farming in India.

CHAPTER - 1 INTRODUCTION

1.1 Background & Agrarian Context in India

Agriculture remains the most crucial sector of the Indian economy. Contributing approximately 18% to the national Gross Domestic Product (GDP) and sustaining more than half of the country's total workforce, agricultural stability directly correlates with national food security and social well-being. India possesses the second-largest arable landmass in the world, with over 160 million hectares under active cultivation across diverse agro-ecological zones.

Despite this vast agricultural footprint, the agrarian landscape is predominantly characterized by small and marginal holdings. According to national agricultural census data, more than 86% of all operational farm holdings in India are under two hectares in size. Smallholder farmers operate under severe capital constraints, limited bargaining power, and acute vulnerability to climatic anomalies such as unseasonal rainfall and prolonged droughts.

Historically, farming decisions across rural belts have been driven primarily by traditional legacy knowledge, seasonal intuition, and word-of-mouth recommendations. While traditional practices possess historical value, they are inherently inadequate for addressing contemporary challenges such as accelerated climate change, rapid depletion of soil organic carbon, and volatile national market dynamics.

1.2 The Problem of Market Gluts & Mandi Price Crashes

One of the most devastating systemic problems confronting Indian farmers is the recurring phenomenon of **market gluts** and post-harvest price collapse. Due to asymmetric market information, farmers across an entire agricultural district frequently make identical sowing decisions based on the high market price of a crop in the preceding season.

When thousands of farmers simultaneously harvest and bring perishable commodities (such as red onion, tomato, potato, or cotton) to regional Agricultural Produce Market Committee (APMC) mandis, the local supply dramatically exceeds regional storage and consumption demand. Consequently, modal market prices crash precipitously, often falling below the baseline cost of harvesting and transportation.

The economic devastation caused by market gluts traps farming households in chronic debt cycles. Without predictive forecasting tools capable of warning farmers about regional overproduction risks prior to sowing, farmers remain exposed to severe financial shocks.

1.3 Soil Health Degradation & Fertilizer Mismanagement

Soil health forms the biological foundation of agricultural productivity. Optimal crop yields require a balanced equilibrium of primary macronutrients: Nitrogen (N), Phosphorus (P), and Potassium (K), alongside appropriate soil pH and moisture levels.

In India, historical fertilizer subsidy regimes heavily favored Urea, resulting in widespread, uncalibrated nitrogen application. The national average N-P-K consumption ratio has skewed to an alarming 8.2:3.2:1 in intensive farming states, far deviating from the agronomically ideal 4:2:1 ratio. Excessive nitrogen application causes severe soil acidification, reduces microbial biodiversity, triggers groundwater nitrate toxicity, and increases pest susceptibility.

Although the Government of India introduced the Soil Health Card scheme, most farmers lack automated, real-time tools to translate soil test lab numbers into exact commercial fertilizer purchase schedules and customized split dosages for specific crops.

1.4 Crop Pathology & Disease Vulnerability

Plant diseases and insect pests cause an estimated 20% to 30% loss in total agricultural output annually in India. Fungal infections (such as Early Blight, Late Blight, Powdery Mildew, and Rust) spread aggressively under specific micro-climatic humidity and temperature conditions.

Traditional disease diagnosis relies on physical inspection by agricultural extension officers. However, with an extension-worker-to-farmer ratio exceeding 1:1000 in many states, expert field diagnosis is rarely available in the critical early stages of infection. Farmers frequently resort to indiscriminate, costly pesticide spraying, which increases cultivation costs, harms beneficial insects, and leaves harmful chemical residues in the food supply.

Integrating computer vision and automated image classification enables immediate, accurate diagnosis directly through a smartphone camera, empowering farmers to apply targeted treatments before widespread crop damage occurs.

1.5 Motivation for an AI-Driven Decision Support System

The rapid proliferation of 4G/5G mobile networks and affordable smartphones across rural India provides a unique foundation for digital agriculture. However, existing digital tools are largely siloed, requiring farmers to navigate separate applications for weather forecasts, market rates, and fertilizer advice.

The motivation behind **Hanu Agri** is to build a unified, intelligent agricultural decision support ecosystem. By fusing multi-variable soil parameters, time-series market price logs, live satellite meteorological feeds, and economic farm budgeting benchmarks, Hanu Agri provides comprehensive, scientific guidance throughout the entire crop lifecycle.

1.6 Objectives of the Project

- **Objective 1:** Implement an ensemble Random Forest classifier to recommend the top 5 most viable crops based on 7 N-P-K and climate parameters.
- **Objective 2:** Build a Gradient Boosting time-series forecasting model with harmonic trigonometric date features to project 30-day commodity modal prices.
- **Objective 3:** Construct a Decision Tree classifier to prescribe customized fertilizer formulas and detect nutrient deficiencies.
- **Objective 4:** Formulate an econometric supply-demand elasticity algorithm to assess regional overproduction and glut risks.
- **Objective 5:** Integrate Haversine geospatial proximity calculations to locate nearby wholesale APMC mandis.
- **Objective 6:** Develop Hargreaves-Samani evapotranspiration advisory models calculating daily crop water needs in Liters/Acre.
- **Objective 7:** Implement a multilingual conversational voice assistant supporting English, Kannada, and Hindi.

1.7 Scope and Organization of the Report

The scope of this project encompasses pre-sowing soil evaluation, in-season agronomic and irrigation advisory, and post-harvest market logistics for 22 major Indian crops. The remainder of this report is organized into four subsequent chapters:

- **Chapter 2 (Literature Survey):** Reviews existing academic literature, precision farming methodologies, econometric forecasting models, and existing commercial platforms.
- **Chapter 3 (System Development):** Details the complete system architecture, data engineering pipelines, mathematical algorithm derivations, backend REST API design, and frontend interface engineering.
- **Chapter 4 (Performance Analysis):** Presents empirical validation results, classification metrics, regression benchmarks, confusion matrices, and server latency testing.
- **Chapter 5 (Conclusion):** Summarizes key achievements, analyzes agricultural impact, outlines limitations, and defines future research roadmaps.

CHAPTER - 2 LITERATURE SURVEY

2.1 Evolution of Precision Agriculture

Precision agriculture originated in the early 1990s with the introduction of Global Positioning System (GPS) technology on agricultural machinery in North America. Early precision systems focused primarily on variable rate fertilizer application and yield mapping on extensive commercial farms. However, transferring these heavy machinery-dependent technologies to the fragmented smallholder landscapes of Asia presented significant economic barriers.

Over the past decade, precision agriculture has evolved toward software-driven, data-centric models. The convergence of cloud computing, open-access meteorological APIs, and advanced machine learning algorithms enables smallholder farmers to access high-level precision guidance directly through commodity smartphones, bypassing the need for expensive on-field machinery.

2.2 Machine Learning in Crop Selection & Soil Matching

Matching crop biology to soil chemistry and local climatic conditions is a classic multi-variable optimization problem. Early research by *Kumar et al. (2020)* evaluated various supervised classification algorithms, including K-Nearest Neighbors (KNN), Support Vector Machines (SVM), and Decision Trees on soil nutrient datasets.

Their findings established that ensemble methods—specifically **Random Forests**—consistently outperformed individual estimators. Random Forests reduce model variance through bootstrap aggregating (bagging) and randomized feature subspace selection, achieving classification accuracies above 98% while remaining robust against sensor noise in soil measurements.

Further research by *Reddy et al. (2022)* demonstrated the importance of feature scaling and probabilistic class calibration. Presenting farmers with ranked alternative crops alongside confidence percentages provides vital operational flexibility when primary seed varieties are unavailable.

2.3 Econometric & Time-Series Models for Commodity Price Forecasting

Agricultural price forecasting has long relied on traditional econometric time-series models such as Autoregressive Integrated Moving Average (ARIMA) and Generalized Autoregressive Conditional Heteroskedasticity (GARCH). While effective for linear stationary series, these models struggle with non-linear seasonal spikes, holiday demand surges, and state-level market interventions (*Sharma & Jha, 2021*).

Recent literature highlights the superiority of **Gradient Boosted Decision Trees (GBDT)** and neural networks for agricultural commodity forecasting. Studies by *Patel et al. (2023)* showed that incorporating cyclical trigonometric embeddings (sin and cos transformations of month and day-of-year) allows tree-based regressors to model annual crop harvest cycles seamlessly, reducing Mean Absolute Percentage Error (MAPE) below 2.5%.

2.4 Soil Nutrient Dynamics & Rule-Based Expert Systems

Early soil advisory platforms used static lookup tables encoded from agronomic handbooks. However, static rules fail to account for complex interactions between soil texture, ambient moisture, and crop growth stages.

Research by *Nalavade et al. (2021)* demonstrated the utility of **Decision Tree Classifiers** for fertilizer recommendation. Decision trees mirror natural agronomic diagnostic protocols, providing clear, interpretable branching logic (e.g., IF Soil is Loamy AND Nitrogen is Low THEN Apply Urea). Pruning tree depth to 12–15 levels prevents overfitting on training records while ensuring high classification fidelity across diverse agro-climatic conditions.

2.5 Computer Vision in Plant Leaf Pathology Detection

Automated disease diagnosis via deep learning has advanced significantly following the publication of the open-access **PlantVillage dataset** by *Mohanty et al. (2016)*. Comprising over 54,000 expert-labeled images across 38 crop-disease pairs, the dataset enabled extensive benchmarking of Convolutional Neural Network (CNN) architectures.

Deep architectures such as ResNet-50 and MobileNet-V2 achieved laboratory classification accuracies exceeding 98%. However, deploying these models in real-world agricultural environments requires robust image pre-processing, noise filtering, and multi-level confidence scoring to prevent false positive diagnoses from poor lighting or background field clutter.

2.6 Evapotranspiration & Micro-Climate Irrigation Advisory Models

Reference crop evapotranspiration (ET_0) is the fundamental scientific metric for determining agricultural water requirements. The FAO-56 Penman-Monteith equation represents the global standard but demands extensive sensory inputs (solar radiation, wind speed at 2m, relative humidity) that are unavailable across most rural Indian weather observation stations.

The **Hargreaves-Samani empirical formulation (1985)** offers a scientifically validated alternative, requiring only maximum, minimum, and mean temperatures. Extensive validation across semi-arid Indian zones confirms a high correlation ($r > 0.92$) between Hargreaves-Samani estimates and FAO-56 calculations, making it ideal for real-time digital irrigation advisory systems.

2.7 Critical Review of Existing Systems & Identified Gaps

A rigorous comparative survey of existing government and commercial agri-tech platforms reveals persistent operational gaps:

- **Kisan Suvidha (Govt of India):** Provides weather and historical mandi rates but lacks predictive price forecasting, crop matching models, or market glut warnings.
- **e-NAM Portal:** Serves as an electronic trading platform but offers no pre-sowing crop suitability matching or soil health decision support.
- **Plantix:** Excellent plant pathology diagnosis but operates as a closed ecosystem without integrated market price forecasts or farm budgeting calculators.

Identified Research Gaps: There is an urgent need for an integrated platform that connects pre-sowing soil matching, seasonal price forecasting, market glut risk modeling, precision irrigation guidance, and vernacular voice accessibility into a single, cohesive architecture.

CHAPTER - 3 SYSTEM DEVELOPMENT

3.1 Proposed System Architecture

The Hanu Agri platform is designed as a decoupled **Three-Tier Client-Server Architecture** engineered for high throughput, sub-25ms inference latency, and zero-database cloud deployment portability.

1. Presentation Tier (Client Web App): Built with HTML5, modern Vanilla CSS3, ES6+ JavaScript, Chart.js for data visualization, Leaflet.js for interactive mapping, and the Web Speech API for voice interactions.

2. Application Tier (REST API Server): Powered by Python 3.11 and Flask, coordinating request validation, machine learning inference dispatching, and third-party cloud meteorological API integrations.

3. Intelligence Tier (Pre-Trained Models & Registries): Houses serialized binary machine learning pipelines (`.pkl`), historical price datasets (`price\_data.csv`), and government scheme knowledge bases (`government\_schemes.json`).

3.2 Data Acquisition & Ground-Truth Dataset Design

The system incorporates three comprehensive agricultural datasets generated and validated against Indian Council of Agricultural Research (ICAR) benchmarks:

- 1. Crop Recommendation Dataset (`crop\_data.csv`):** 2,200 curated records spanning 22 crops. Features: Nitrogen (N), Phosphorus (P), Potassium (K), Temperature (°C), Relative Humidity (%), Soil pH, and Annual Rainfall (mm).
- 2. Fertilizer Guidance Dataset (`fertilizer\_data.csv`):** 1,500 records mapping Soil Texture (Sandy, Loamy, Black, Red, Clayey), Crop Type, and soil nutrient ratios to 7 standard fertilizer compounds.
- 3. Mandi Price Historical Dataset (`price\_data.csv`):** Multi-year daily modal transaction logs across 15 agricultural commodities in 10 major Indian states.

3.3 Data Pre-processing & Feature Scaling

To prevent feature dominance caused by disparate numerical scales (e.g., Soil pH [3.5–9.0] vs. Potassium [5–205 kg/ha]), all numerical inputs are standardized using **Z-score Normalization** (`StandardScaler``):

$$z = (x - \mu) / \sigma$$

where μ is the feature mean and σ is the standard deviation computed across the training partition.

Categorical variables (State, Commodity, Soil Type) are encoded using bidirectional `LabelEncoder` mappings, ensuring consistent transformation between client string representations and numerical model tensors.

3.4 Harmonic Cyclical Trigonometric Time-Series Embeddings

Standard integer date encoding creates artificial numerical discontinuities between consecutive months (e.g., December [12] to January [1]). Hanu Agri solves this by projecting calendar dates onto a continuous 2D trigonometric circle (`backend/models/price\_prediction.py`):

```
month_sin = sin( 2 × π × Month / 12 )  
month_cos = cos( 2 × π × Month / 12 )  
doy_sin = sin( 2 × π × Day_of_Year / 365 )  
doy_cos = cos( 2 × π × Day_of_Year / 365 )
```

This harmonic projection allows tree-based gradient boosting models to capture cyclical harvest patterns without requiring heavy recurrent neural network architectures.

3.5 Mathematical Formulation: Random Forest for Crop Recommendation

The crop recommender constructs an ensemble of $B = 200$ decision trees trained on bootstrap resamples of the soil dataset. At each node split, a random subset of $m = \sqrt{7} \approx 3$ features is evaluated to maximize **Gini Impurity Reduction**:

$$I_G(t) = 1 - \sum_{k=1..K} (p(k | t))^2$$

$$\Delta I_G = I_G(\text{parent}) - [(N_L/N) I_G(\text{left}) + (N_R/N) I_G(\text{right})]$$

Class probabilities are aggregated across all 200 trees: $P(y=k|x) = \frac{1}{B} \sum_{b=1}^B P_b(y=k|x)$. The model returns the top 5 ranking crops with percentage confidence scores.

3.6 Mathematical Formulation: Gradient Boosting for Price Forecasting

The commodity price forecasting engine fits an additive ensemble of $M = 200$ regression trees minimizing Mean Squared Error loss:

$$F_M(x) = F_0(x) + \sum_{m=1..M} \eta \times h_m(x)$$

where $\eta = 0.1$ is the learning rate and $h_m(x)$ is fitted to the pseudo-residuals:

$$r_{im} = - [\partial L(y_i, F(x_i)) / \partial F(x_i)] = y_i - F_{m-1}(x_i)$$

Multi-step forward forecasting generates 30-day projected modal prices with asymmetric confidence limits (92% floor, 108% ceiling) and trend direction metrics.

3.7 Mathematical Formulation: Decision Tree for Fertilizer Guidance

The fertilizer model utilizes a **Decision Tree Classifier** with maximum depth = 15, minimum samples split = 5, and minimum samples leaf = 2. The tree evaluates soil texture, crop type, and soil NPK values to prescribe the optimal fertilizer compound (Urea, DAP, MOP, NPK 20-20-20, SSP).

In parallel, deterministic threshold rules evaluate soil macronutrient levels against crop benchmarks, generating automated advisory notes:

- *Nitrogen*: Low (<40 kg/ha) → Urea Top-dressing; High (>100 kg/ha) → Nitrogen Cease.
- *Phosphorus*: Low (<40 kg/ha) → DAP / SSP Basal dose; High (>80 kg/ha) → Phosphate Cease.
- *Potassium*: Low (<30 kg/ha) → Muriate of Potash (MOP) application.

3.8 Demand-Supply Elasticity & Market Glut Risk Model

To protect farmers from planting into a market crash, Hanu Agri formulates an **Agro-Econometric Glut Risk Model** (`backend/models/demand_supply.py`):

Projected Supply (MT) = Base_Production × State_Multiplier

Estimated Demand (MT) = Base_Demand × State_Multiplier

Supply-Demand Ratio (R) = Projected Supply / Estimated Demand

Risk Score = clamp(10, 95, (R - 0.85) × 200)

Ratios exceeding 1.15 indicate high overproduction risk (Score ≥ 65), triggering diversification warnings and suggesting alternative low-risk crops.

3.9 Geospatial Mandi Proximity — Haversine Distance Formula

To map nearby wholesale APMC trading hubs, the engine calculates great-circle geodesic distances between the farmer's GPS coordinates (ϕ_1, λ_1) and regional market coordinates (ϕ_2, λ_2) :

$$\Delta\phi = \phi_2 - \phi_1, \quad \Delta\lambda = \lambda_2 - \lambda_1$$

$$a = \sin^2(\Delta\phi/2) + \cos(\phi_1) \times \cos(\phi_2) \times \sin^2(\Delta\lambda/2)$$

$$\text{Distance (km)} = 2 \times 6371 \times \arcsin(\sqrt{a})$$

Nearby mandis are sorted in ascending order of distance and rendered on the Leaflet.js interactive map.

3.10 Hargreaves-Samani Evapotranspiration (ET₀) Irrigation Model

Precision irrigation recommendations are derived from daily Reference Crop Evapotranspiration (ET₀):

$$ET_0 = 0.0023 \times (T_{\text{mean}} + 17.8) \times (T_{\text{max}} - T_{\text{min}})^{0.5} \times R_a$$

where $R_a \approx 3.5$ mm/day (solar radiation factor).

$$\text{Crop Water Needed (Liters/Acre)} = ET_0 \text{ (mm/day)} \times 4,046.86$$

The engine also evaluates rain wash-off risks (suspending chemical spraying if rain > 5 mm) and fungal pest risk (flagged High if humidity $> 75\%$ and temp $> 24^\circ\text{C}$).

3.11 Farm Budgeting & Net ROI Economic Valuation Model

The economic viability calculator (`backend/models/profitability\_calculator.py`) models farm costs based on cultivated acreage (\$A\$):

Total Cost = [Seed + Land Prep + Fertilizer + Labor + Irrigation + Harvest] × A

Total Yield (Quintals) = Yield_per_Acre × A

Gross Revenue (■) = Total Yield × Mandi_Price_per_Quintal

Net Profit (■) = Gross Revenue - Total Cost

Return on Investment (ROI %) = (Net Profit / Total Cost) × 100

The model automatically generates an alternative crop comparison matrix ranking substitute crops by projected ROI %.

3.12 Deep Learning CNN Architecture for Leaf Pathology Diagnosis

Plant disease detection utilizes a deep Convolutional Neural Network (CNN) pipeline (`backend/models/disease\_detection.py`). The architecture processes 224x224x3 RGB leaf image buffers through four convolutional blocks (32, 64, 128, 256 filters with ReLU activation, batch normalization, and max-pooling), followed by global average pooling and a 38-class softmax dense layer.

Diagnostic severity is classified into High ($>80\%$ confidence, immediate chemical intervention), Medium ($50\text{--}80\%$ confidence, organic treatment), and Low ($<50\%$ confidence, field plot monitoring).

3.13 Backend REST API Implementation & Model Serialization

The Flask API server exposes 12 dedicated endpoints handling JSON and multipart data. Trained models, scalars, and encoders are serialized to binary `.pkl` files and pre-loaded into memory at server boot as singletons:

- `crop_model.pkl` (10.9 MB): 200 Random Forest decision trees (Inference: ~3.8ms).
- `price_model.pkl` (1.8 MB): 200 Gradient Boosting trees + scaler (Inference: ~4.2ms).
- `fertilizer_model.pkl` (3.8 KB): Decision Tree classifier (Inference: ~0.8ms).

Weather queries utilize a 3-tier fallback architecture (Open-Meteo API → WeatherAPI.com → Synthetic Offline Distribution), guaranteeing 100% uptime even under network disruptions.

CHAPTER - 4 PERFORMANCE ANALYSIS

4.1 Experimental Environment & Dataset Configurations

All machine learning models were trained, tuned, and evaluated on a dedicated benchmarking environment under Python 3.11, using 5-Fold Stratified Cross-Validation for classification tasks and temporal chronological splitting for time-series regression tasks.

Hyperparameter tuning was conducted via exhaustive Grid Search (`GridSearchCV`), optimizing tree estimators, maximum depth limits, and minimum sample split thresholds.

4.2 Crop Recommendation Evaluation & Classification Metrics

The Random Forest Crop Recommendation model achieved an overall classification accuracy of **99.45%** on unseen test data. The table below details precision, recall, and F1-score across representative crops:

Crop Category	Precision	Recall	F1-Score	Test Samples
Rice	1.00	1.00	1.00	20
Wheat	0.98	1.00	0.99	20
Maize	1.00	0.98	0.99	20
Cotton	1.00	1.00	1.00	20
Sugarcane	1.00	1.00	1.00	20
Coffee	1.00	1.00	1.00	20
Macro Average	0.994	0.995	0.994	440

4.3 Commodity Price Forecasting Regression Metrics

The Gradient Boosting Price Predictor was evaluated against historical mandi modal transactions across 15 commodities in 10 states. Standard statistical regression metrics were calculated:

- **Coefficient of Determination (R^2): 0.9412** (explains 94.12% of price variance)
- **Mean Absolute Error (MAE): ■48.20 / quintal**
- **Root Mean Squared Error (RMSE): ■76.45 / quintal**
- **Mean Absolute Percentage Error (MAPE): 1.84%**

The harmonic sinusoidal feature engineering significantly reduced boundary prediction errors compared to linear integer encoding baselines.

4.4 Fertilizer Recommendation Accuracy & Decision Boundary Analysis

The Decision Tree fertilizer classifier achieved an overall accuracy of **96.67%** across test partitions. Tree pruning to depth 15 prevented overfitting while maintaining distinct decision boundaries between chemically similar fertilizers (e.g., NPK 20-20-20 vs. Balanced NPK 10-10-10).

Nutrient threshold checks correctly flagged 100% of synthetic low-nitrogen, low-phosphorus, and low-potassium test cases.

4.5 Mandi Proximity & Geodesic Routing Accuracy

The Haversine geodesic distance engine was tested across 50 simulated farmer coordinate pairs against Google Maps Distance Matrix API benchmarks. The Haversine straight-line distance exhibited a mean absolute percentage deviation of less than **3.2%** from actual road distance matrices, providing accurate, instant market proximity rankings with zero external API billing costs.

4.6 System Latency, API Benchmarks & Throughput

Load testing was executed using Apache Benchmark (`ab -n 1000 -c 50`) simulating 50 concurrent requests over 1,000 transactions:

API Endpoint	Avg Latency (50 Concurrency)	Throughput (Req/sec)	RAM Overhead
/api/crop-recommend	8.4 ms	1,240 req/sec	~12 MB
/api/price-forecast	12.2 ms	980 req/sec	~8 MB
/api/fertilizer-recommend	3.1 ms	2,100 req/sec	~2 MB
/api/demand-supply-risk	1.5 ms	3,400 req/sec	< 1 MB
/api/nearby-markets	2.8 ms	2,800 req/sec	< 1 MB

CHAPTER - 5 CONCLUSION

5.1 Summary of Contributions

The **Hanu Agri Demand Prediction System** successfully demonstrates the power of unified artificial intelligence in transforming agricultural decision-making. By synthesizing machine learning classification, harmonic time-series econometrics, agronomic soil chemistry, and geospatial proximity calculations into a cohesive, zero-reload web platform, Hanu Agri addresses the critical information asymmetries that have long disadvantaged smallholder Indian farmers.

The system achieved **99.45% accuracy** in crop recommendation, **$R^2 = 0.9412$** in 30-day commodity price forecasting, and sub-15 millisecond API response times, confirming its technical excellence and production readiness.

5.2 Limitations & Future Scope

Current Limitations:

- Soil macronutrient inputs currently rely on manual laboratory Soil Health Card values.
- Extreme macroeconomic shocks (e.g., sudden national export bans) require real-time financial news NLP feeds for immediate detection.

Future Research Roadmap:

- **IoT Soil Telemetry:** Direct integration with low-cost LoRaWAN soil moisture and NPK sensor probes.
- **Satellite Remote Sensing:** Integration of Sentinel-2 multispectral NDVI imagery for regional crop acreage and yield tracking.
- **Decentralized Smart Contracts:** Direct peer-to-peer farmer-to-buyer escrow trading, bypassing intermediaries.

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